

AI Resume Ranker - Project Report

Introduction

AI Resume Ranker is a Flask-based web application that ranks multiple resumes against a job description using Natural Language Processing (NLP). It automates the initial resume screening process by calculating similarity scores.

Abstract

The project extracts text from PDF resumes, converts the job description and resumes into TF-IDF vectors, computes cosine similarity, and ranks candidates based on relevance. This reduces manual effort and improves shortlisting efficiency.

Tools Used

Python, Flask, PyPDF2, Scikit-learn (TF-IDF Vectorizer and Cosine Similarity), HTML, Git, GitHub, VS Code, Anaconda.

Steps Involved in Building the Project

1. Set up the Python environment and install required libraries. 2. Develop the Flask application. 3. Create the HTML interface for uploading resumes and entering a job description. 4. Extract text from PDF files using PyPDF2. 5. Convert text into TF-IDF vectors. 6. Calculate cosine similarity between the job description and each resume. 7. Rank resumes by similarity score and display the results. 8. Test the application and upload the project to GitHub.

Conclusion

The AI Resume Ranker demonstrates how NLP can automate resume screening. The project provides a simple, efficient, and accurate way to rank resumes, making it useful as an academic project and a foundation for more advanced recruitment systems.